

Mathematics I: Reference Notes

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1 Algebra

1.1 Expressions and brackets

1.1.1 The language of algebra

Definition 1.1 Expression

An **expression** is an arrangement of letters and numbers joined by mathematical symbols, such that substituting numbers for any letters gives a value that can be worked out. For example, $5y - 2$ is an expression, but $5y - \times 2$ is not.

An expression containing letters is an **algebraic expression**. One containing only numbers and symbols, such as $7 + 3 \times 2$, is a **numerical expression**. A **subexpression** is part of a larger expression that can be enclosed in brackets without changing its meaning. For example, $5y$ is a subexpression of $5y - 2$, but $y - 2$ is not, since $5(y - 2)$ means something different.

Definition 1.2 Types of letter

A **variable** represents any number, or any number of a particular kind, such as any positive number or any integer. An **unknown** represents a particular number that is not known, usually to be found by solving an equation. A **constant** is a particular number whose value is specified, or fixed for a calculation, such as $\pi \approx 3.14$.

In practice all letters that represent numbers are treated in the same way, and “variable” is often used for both variables and unknowns.

Multiplication signs are usually omitted unless they sit between two numbers, so $7 \times t$ is written $7t$. Division is written as a fraction, so $7 \div t$ is written $\frac{7}{t}$, or $7/t$ within a line of text.

Definition 1.3 Equation

Expressions do not contain equals signs. An **equation** is made up of two expressions with an equals sign between them, such as $4y - y = 3y$.

An expression made of quantities that are added or subtracted is best thought of as quantities that are all added. For example,

$$4pq - 3q^2 + 7p + 2\sqrt{q} - 9 = 4pq + (-3q^2) + 7p + 2\sqrt{q} + (-9).$$

Definition 1.4 Terms and coefficients

The quantities that are added are the **terms** of the expression. A term with no variables is a **constant term**. If a term has the form (fixed value) $\times$ (combination of variables), the fixed value is the **coefficient** of the term, and the term is a term *in* that combination of variables.

In the expression above, $4pq$ has coefficient 4 and is a term in pq , $-3q^2$ has coefficient -3 and is a term in q^2 , $2\sqrt{q}$ has coefficient 2 and is a term in $\sqrt{q}$, and -9 is a constant term.

A coefficient can be 1 or -1 : the terms of $y - y^3$ have coefficients 1 and -1 . A coefficient can also include constants: in $5r^2 - 2\pi r^2$, the second term is a term in r^2 with coefficient -2π .

Definition 1.5 Equivalent expressions

Two expressions are **equivalent** if they have the same value whatever values are chosen for their variables. This is shown with an equals sign (some texts use $\equiv$). Either expression can be **rearranged** or **rewritten** to give the other. **Simplifying** an expression means rewriting it as a simpler equivalent expression.

Changing the order of the terms gives an equivalent expression, since the order of addition does not affect the result:

$$4pq - 3q^2 + 7p + 2\sqrt{q} - 9 = 7p - 9 + 2\sqrt{q} - 3q^2 + 4pq.$$

1.1.2 Simplifying terms and expressions**Definition 1.6** Collecting like terms

Terms that differ only in their coefficients are **like terms**. They can be combined into a single term by adding their coefficients, which is called **collecting like terms**.

For example,

$$4k - 7k + 10k = 7k, \quad 2m^2 + mn - 6m^2 + 4mn = -4m^2 + 5mn.$$

Individual terms can also be simplified, for example $uv(-v) = -uv^2$.

Proposition 1.7 Strategy for simplifying a term

1. Find the overall sign and write it at the front.
2. Simplify the rest of the coefficient and write it next.
3. Write the remaining parts of the term in a sensible order, usually alphabetical, using index notation to avoid repeating letters.

Proposition 1.8 Sign rules

When multiplying or dividing, two signs the same give a plus sign and two different signs give a minus sign. Adding the negative of something is the same as subtracting it, and subtracting the negative of something is the same as adding it.

A plus or minus sign at the start of a term has the same effect as multiplying the term by 1 or -1 , so the sign rules apply to these signs too.

Working example tidying up individual terms

(a) $+(-5k) = -5k$

(b) $-(-3m) = 3m$

(c) The overall sign is plus, the rest of the coefficient is $4 \times 5 = 20$, and $xy^2 \times x^3y = x^4y^3$, so

$$-4xy^2 \times (-5x^3y) = 20x^4y^3.$$

(d) There are three minus signs, so the overall sign is minus:

$$-m \times n \times (-m) \times (-n) \times n = -m^2 n^3.$$

(e) $(3y)^3 = (3y) \times (3y) \times (3y) = 27y^3$

Proposition 1.9 Index law for multiplication

For any natural numbers m and n ,

$$a^m \times a^n = a^{m+n}.$$

For example, $a^4 \times a^2 = (a \times a \times a \times a) \times (a \times a) = a^6$.

Proposition 1.10 Strategy for simplifying an expression with more than one term

1. Identify the terms. Each term after the first starts with a plus or minus sign that is not inside brackets.
2. Simplify each term, including the sign at its start (a plus sign on the first term can be dropped).
3. Collect like terms.

Working example simplifying a multi-term expression

$$\begin{aligned} \underbrace{4a \times 3b} \quad \underbrace{-2a(-a)} \quad \underbrace{-5a^2} &= 12ab + 2a^2 - 5a^2 \\ &= 12ab - 3a^2 \end{aligned}$$

This could also be written $-3a^2 + 12ab$, but $12ab - 3a^2$ is slightly tidier.

1.1.3 Expanding brackets

Definition 1.11 Expanding brackets

Rewriting an expression such as $2m(4 - 3n)$ as an expression without brackets is called **multiplying out**, **expanding** or **removing** the brackets. The result is the **expansion** of the original expression. The subexpression that multiplies the brackets, here $2m$, is the **multiplier**.

Proposition 1.12 Strategy for multiplying out brackets

Multiply each term inside the brackets by the multiplier.

For example,

$$2m(4 - 3n) = 2m \times 4 + 2m \times (-3n) = 8m - 6mn.$$

Every term inside the brackets must be multiplied, not just the first. It is tidier to simplify each term mentally as it is multiplied out, rather than writing down the multiplication signs first. The multiplier can also come after the brackets, as in $(3k + 2)j = 3jk + 2j$.

Working example expanding a single pair of brackets

The terms of the expansion are $-3a \times a$, $-3a \times (-2b)$ and $-3a \times 4$, each simplified by finding the overall sign, then the coefficient, then the letters:

$$-3a(a - 2b + 4) = -3a^2 + 6ab - 12a.$$

Brackets preceded by a sign. A minus sign in front of brackets is the same as multiplying by -1 , and a plus sign is the same as multiplying by 1 .

Proposition 1.13 Strategy for removing brackets with a plus or minus sign in front

If the sign is plus, keep the sign of each term inside the brackets. If the sign is minus, change the sign of each term inside the brackets.

For example, $-(2m - n + 3k) = -2m + n - 3k$ and $+(x + 4y) = x + 4y$.

Proposition 1.14 Negative of a difference

For any expressions A and B ,

$$-(A - B) = B - A.$$

For example, $-(p - q) = q - p$ and $-(2t - t^3) = t^3 - 2t$.

Proposition 1.15 Multiplying and dividing by an expression

Multiplying an expression with several terms by a second expression is the same as multiplying *each term* of the first expression by the second. Dividing works in the same way, since dividing by z is multiplying by $1/z$:

$$\frac{x + 3y - 5}{z} = \frac{1}{z}(x + 3y - 5) = \frac{x}{z} + \frac{3y}{z} - \frac{5}{z}.$$

Expanding brackets within longer expressions.

Proposition 1.16 Strategy for multiplying out brackets in an expression with more than one term

1. Identify the terms. Each term after the first starts with a plus or minus sign that is not inside brackets.
2. Multiply out the brackets in each term, including the sign at the start of each resulting term.
3. Collect like terms.

Working example expanding brackets in a longer expression

The expression has three terms, two of which contain brackets. Multiplying these out gives five terms, which are then collected.

$$\underbrace{3m(m-n)} + \underbrace{4mn} - \underbrace{n(2m+n)} = 3m^2 - 3mn + 4mn - 2mn - n^2 \\ = 3m^2 - mn - n^2$$

Products of two pairs of brackets. Choose one of the bracketed expressions to be the multiplier and multiply out the other pair of brackets. This gives several terms that each contain a pair of brackets, which are then multiplied out as usual. Either choice works, but taking the second pair as the multiplier is usually slightly easier.

Working example expanding a product of two brackets

Take $a + 3b$ as the multiplier.

$$(a + 2ab - b)(a + 3b) = a(a + 3b) + 2ab(a + 3b) - b(a + 3b) \\ = a^2 + 3ab + 2a^2b + 6ab^2 - ab - 3b^2 \\ = a^2 + 2ab + 2a^2b + 6ab^2 - 3b^2$$

The effect is that each term in the first pair of brackets is multiplied by each term in the second, and the results are added. Three terms multiplied by two terms gives $3 \times 2 = 6$ terms before like terms are collected.

Proposition 1.17 FOIL

When each pair of brackets contains two terms, there are four multiplications: the **F**irst terms, the **O**uter terms, the **I**nnner terms and the **L**ast terms.

Working example applying FOIL

For $(2y - 3)(y + 4)$:

$$\text{F: } 2y \times y = 2y^2, \quad \text{O: } 2y \times 4 = 8y, \quad \text{I: } -3 \times y = -3y, \quad \text{L: } -3 \times 4 = -12.$$

So

$$(2y - 3)(y + 4) = 2y^2 + 8y - 3y - 12 = 2y^2 + 5y - 12.$$

A sum multiplied by a difference. When the two pairs of brackets differ only in the sign between their terms, the outer and inner products add to zero. For example,

$$(y + 4)(y - 4) = y^2 - 4y + 4y - 16 = y^2 - 16.$$

This always happens for an expression of the form $(A + B)(A - B)$.

Proposition 1.18 Difference of two squares

For any expressions A and B ,

$$(A + B)(A - B) = A^2 - B^2.$$

An expression of the form $A^2 - B^2$ is called a **difference of two squares**.

Squaring a single pair of brackets. Squared brackets can be multiplied out by writing them as two pairs of brackets multiplied together:

$$(y + 3)^2 = (y + 3)(y + 3) = y^2 + 3y + 3y + 9 = y^2 + 6y + 9.$$

Note that $(y + 3)^2$ is *not* equal to $y^2 + 3^2$.

Proposition 1.19 Squaring brackets

For any expressions A and B ,

$$(A + B)^2 = A^2 + 2AB + B^2, \quad (A - B)^2 = A^2 - 2AB + B^2.$$

Working example squaring brackets in one step

- (a) The square of $2a$, plus twice the product of $2a$ and b , plus the square of b :

$$(2a + b)^2 = (2a)^2 + 2 \times (2a) \times b + b^2 = 4a^2 + 4ab + b^2.$$

- (b) The square of $5k$, minus twice the product of $5k$ and 2 , plus the square of 2 :

$$(5k - 2)^2 = (5k)^2 - 2 \times (5k) \times 2 + 2^2 = 25k^2 - 20k + 4.$$

Choosing a simplest form. A simpler form of an expression is shorter and easier to evaluate, so $y + 4y - 2y$ should be written as $3y$. Often there is no single simplest form. For example, $y(y - 2)$ and $y^2 - 2y$ are both short and easy to evaluate, and one may suit a particular purpose better than the other. Multiplying out brackets does not always simplify an expression, so only do it when it makes the expression simpler or helps with the problem at hand.

1.2 Factorising and fractions

Algebraic expressions have factors and multiples in the same way that integers do. For example,

$$xy^2 = y \times xy$$

shows that y and xy are both factors of xy^2 , and that xy^2 is a multiple of both y and xy . Every expression is a factor and a multiple of itself, since $xy^2 = 1 \times xy^2$.

Two or more expressions can also share common factors and common multiples. For example, y is a common factor of xy^2 and y^3z , and pqr is a common multiple of pq and qr because

$$pqr = pq \times r \quad \text{and} \quad pqr = qr \times p.$$

To show that an expression is a common factor, write each expression as that factor multiplied by something. For example, $4w$ is a common factor of $12w^3$, $8w^2$ and $4w$ because

$$12w^3 = 4w \times 3w^2, \quad 8w^2 = 4w \times 2w, \quad 4w = 4w \times 1.$$

1.2.1 Finding the HCF and LCM of expressions

Definition 1.20 HCF and LCM

Expressions with integer coefficients have an HCF and an LCM, just as integers do. The **HCF** is a common factor that is a multiple of every other common factor. The **LCM** is a common multiple that is a factor of every other common multiple.

Working example HCF and LCM of $12p^3q^2$ and $8p^5$

HCF. Work through each part of the expressions in turn. The largest integer that is a factor of both 12 and 8 is 4. The largest power of p that is a factor of both p^3 and p^5 is p^3 , since

$$p^3 = p^3 \times 1 \quad \text{and} \quad p^5 = p^3 \times p^2.$$

There is no q in $8p^5$, so the HCF contains no power of q . The HCF is $4p^3$, and

$$12p^3q^2 = 4p^3 \times 3q^2, \quad 8p^5 = 4p^3 \times 2p^2.$$

LCM. The smallest positive integer that is a multiple of both 12 and 8 is 24. The smallest power of p that is a multiple of both p^3 and p^5 is p^5 . The smallest power of q that is a multiple of both q^2 and no power of q is q^2 . The LCM is $24p^5q^2$, and

$$24p^5q^2 = 12p^3q^2 \times 2p^2, \quad 24p^5q^2 = 8p^5 \times 3q^2.$$

1.2.2 Factorising by extracting a common factor

Definition 1.21 Factorising

To **factorise** an expression means to write it as a product of two or more expressions, usually none of which is 1 or -1 .

For example, the terms of $a^2b^3 - ab$ have ab as a common factor, so

$$\begin{aligned} a^2b^3 - ab &= ab \times ab^2 - ab \times 1 \\ &= ab(ab^2 - 1). \end{aligned}$$

Proposition 1.22 Strategy for factorising an expression

1. Find the common factor of the terms (the HCF).
2. Write the common factor in front of a pair of brackets.
3. Write what is left of each term inside the brackets.

Taking out a negative factor. Sometimes it is convenient to take out the common factor with a minus sign. For example,

$$-6k^4 + 9k^2m - 3k^2 = -3k^2(2k^2 - 3m + 1).$$

Taking out a minus sign reverses the sign of every term inside the brackets, as in

$$-x + y - z = -(x - y + z).$$

Factorising with fractional coefficients. If the coefficients of the terms are not integers, the expression can still be factorised. When the coefficients are fractions, the denominator of the fraction taken out is typically a common multiple of the denominators of the coefficients.

Working example extracting a fractional factor

$$\frac{3}{4}t^2 + \frac{1}{2}t - 2 = \frac{1}{4}(3t^2 + 2t - 8)$$

Here 4 is a common multiple of the denominators 4 and 2. Multiplying back out confirms each term:

$$\frac{1}{4} \times 3 = \frac{3}{4}, \quad \frac{1}{4} \times 2 = \frac{2}{4} = \frac{1}{2}, \quad \frac{1}{4} \times (-8) = -\frac{8}{4} = -2.$$

Proposition 1.23 Common factor of fractions

The common factor of fractional coefficients is

$$\frac{\text{HCF of the numerators}}{\text{LCM of the denominators}}.$$

For example, for $\frac{4}{5}$ and $\frac{2}{3}$ this is $\frac{\text{HCF}(4, 2)}{\text{LCM}(5, 3)} = \frac{2}{15}$.

Sometimes the expression in the brackets has further factors that can be taken out.

1.2.3 Reducing algebraic fractions to lowest terms

A numerical fraction is simplified by dividing its numerator and denominator by a common factor:

$$\frac{6}{9} = \frac{\cancel{6}^2}{\cancel{9}_3} = \frac{2}{3}.$$

The same can be done for algebraic fractions. For example, b is a common factor of b^4 and ab , so

$$\frac{b^4}{ab} = \frac{\cancel{b}^3}{a\cancel{b}_1} = \frac{b^3}{a}.$$

Definition 1.24 Cancelling

Dividing the numerator and denominator of a fraction by a common factor is called **cancelling**, or **cancelling down**.

Working example dividing top and bottom by the HCF

Find the HCF of the numerator and denominator, then divide both by it.

(a) The HCF of $10m^3n$ and $15mn^2$ is $5mn$, so

$$\frac{10m^3n}{15mn^2} = \frac{2m^2}{3n}.$$

(b) The HCF of $y^2(y-3)^2$ and $y^4(y-3)$ is $y^2(y-3)$, so

$$\frac{y^2(y-3)^2}{y^4(y-3)} = \frac{y-3}{y^2}.$$

In some cases there won't be any obvious common factors. Factorising the numerator and denominator can then create some.

Working example factorising first to reveal a common factor

(a) Factorising gives a as a common factor:

$$\frac{3a^2 - a}{a^3 + 4a} = \frac{a(3a - 1)}{a(a^2 + 4)} = \frac{3a - 1}{a^2 + 4}.$$

(b) Factorising gives $v + 3$ as a common factor:

$$\frac{4v + 12}{v^2 + 3v} = \frac{4(v + 3)}{v(v + 3)} = \frac{4}{v}.$$

1.2.4 Sums and differences of algebraic fractions

With numerical fractions, the denominators need to be the same before the fractions can be added or subtracted. The same method can be used for algebraic fractions.

Proposition 1.25 Adding and subtracting fractions

1. Make sure the fractions have the same denominator, rewriting them as equivalent fractions if needed.
2. Add or subtract the numerators.
3. Simplify if possible.

Working example finding a common denominator

- (a) The denominators are already the same:

$$\frac{2y+5}{y^3} - \frac{5}{y^3} = \frac{2y}{y^3} = \frac{2}{y^2}.$$

- (b) Multiplying the top and bottom of the second and third fractions makes every denominator pq :

$$\frac{3}{pq} - \frac{1}{p} + \frac{4}{q} = \frac{3}{pq} - \frac{q}{pq} + \frac{4p}{pq} = \frac{3-q+4p}{pq}.$$

- (c) Here the two denominators are multiplied by different things to make them the same, and each numerator is multiplied by the same term as its own denominator:

$$\begin{aligned}\frac{t}{t+2} - \frac{3}{t} &= \frac{t^2}{t(t+2)} - \frac{3(t+2)}{t(t+2)} \\ &= \frac{t^2 - 3(t+2)}{t(t+2)} = \frac{t^2 - 3t - 6}{t(t+2)}.\end{aligned}$$

- (d) A whole term can be written over 1, then multiplied top and bottom by m to match the other denominator:

$$k - \frac{2}{m} = \frac{k}{1} - \frac{2}{m} = \frac{km}{m} - \frac{2}{m} = \frac{km-2}{m}.$$

1.2.5 Splitting a fraction into separate terms

When an expression is a sum of several terms, dividing it by a second expression is the same as dividing each term of the first expression by the second. For example,

$$\frac{3a^3 + b - 4}{a} = \frac{3a^3}{a} + \frac{b}{a} - \frac{4}{a} = 3a^2 + \frac{b}{a} - \frac{4}{a}$$

This is called **expanding** the fraction and is essentially the reverse procedure to adding or subtracting fractions shown above.

Algebraic fractions can only be expanded if there is a sum of terms in the **numerator**. The following fraction can't be expanded.

$$\frac{a}{3a^3 + b - 4}$$

1.2.6 Products and quotients of algebraic fractions

Algebraic fractions can be multiplied and divided in the same way as numerical fractions. For example:

$$\frac{4a}{3b} \times \frac{6b^3}{5a^2} = \frac{4a \times 6b^3}{3b \times 5a^2} = \frac{24ab^3}{15a^2b} = \frac{8b^2}{5a}$$

Dividing algebraic fractions is done by multiplying the first fraction by the **reciprocal** of the second. For example:

$$\frac{4a}{3b} \div \frac{6b^3}{5a^2} = \frac{4a}{3b} \times \frac{5a^2}{6b^3} = \frac{4a \times 5a^2}{3b \times 6b^3} = \frac{20a^3}{18b^4} = \frac{10a^3}{9b^4}$$

Definition 1.26 Cross cancelling

When multiplying or dividing fractions, any common factors in the numerator and denominator can be cancelled before multiplying or dividing. This is called **cross cancelling**.

An example of cross cancelling is shown below:

Working example cancelling before multiplying

(a) Cancel a factor of $y + 3$ before multiplying:

$$\frac{y+3}{5} \times \frac{y-2}{(y+3)^3} = \frac{\cancel{y+3}}{5} \times \frac{y-2}{\cancel{(y+3)^3}^{(y+3)^2}} = \frac{y-2}{5(y+3)^2}.$$

(b) Multiply by the reciprocal, then cancel before multiplying:

$$\begin{aligned} \frac{8M^5}{N^2} \div \frac{2M^2}{N^6} &= \frac{8M^5}{N^2} \times \frac{N^6}{2M^2} \\ &= \frac{\overset{4}{\cancel{8}} \overset{M^3}{\cancel{M^5}}}{\cancel{N^2}^1} \times \frac{\overset{N^4}{\cancel{N^6}}^2}{\overset{2}{\cancel{M^2}}^1} = 4M^3N^4. \end{aligned}$$

1.3 Surds and indices

Working with roots, powers of numbers, and algebraic expressions.

1.3.1 Square, cube and higher roots

Definition 1.27 Square root

A **square root** of a number is a number that, when raised to the power of 2, gives the original number. For example, 5 and -5 are square roots of 25 because $5^2 = 25$ and $(-5)^2 = 25$.

Definition 1.28 Cube root

A **cube root** of a number is a number that, when raised to the power of 3, gives the original number. For example, 3 is a cube root of 27 because $3^3 = 27$.

Fourth roots, fifth roots, and so on can be defined in a similar way. The n th root of a number x is a number y such that $y^n = x$.

Definition 1.29 Negative numbers

Negative numbers do not have real square roots. For example, there is no real number x such that $x^2 = -4$. However, negative numbers **do have real cube roots**. For example, -3 is a cube root of -27 because $(-3)^3 = -27$.

For all **even roots**, the root of a negative number is not a real number. For all **odd roots**, the root of a negative number is a negative number.

You can use the $\pm$ symbol to indicate that a number has two roots. For example, the square roots of 16 are ± 4 , which means that both 4 and -4 are square roots of 16.

Proposition 1.30 Using $\pm$ and $\sqrt{x}$

Because $\sqrt{x}$ is used to denote a positive square root, we can write $\sqrt{16} = 4$. To indicate both roots, we can use $\pm$; for example, if $x^2 = 36$ then $x = \pm\sqrt{36} = \pm 6$.

The number to the upper left of the root symbol denotes the n th root. For example, $\sqrt[3]{64} = 4$ because $4^3 = 64$. The cube root of -64 is -4 because $(-4)^3 = -64$.

Most square roots are irrational numbers. For example, $\sqrt{3}$ is an irrational number because it cannot be expressed as a fraction of two integers. However, some square roots are rational numbers, such as $\sqrt{9} = 3$. **They are often just left in root form, such as $\sqrt{3}$, because they cannot be simplified.**

Definition 1.31 Surds

An expression such as $3 + \sqrt{7}$ is called a **surd**. A surd is an expression that contains a root that cannot be simplified to remove the root. For example, $\sqrt{3}$ is a surd because it cannot be simplified to remove the square root. However, $\sqrt{9}$ is not a surd because it can be simplified to 3.

Some common surds:

$$\sqrt{2} \quad \sqrt{3} \quad \sqrt{5} \quad \sqrt{7} \quad \sqrt{11} \quad \sqrt{13}$$

1.3.2 Surd arithmetic

You can manipulate surds using the normal rules of algebra and treat them the same way you would a variable, for example when collecting like terms.

Such as:

$$2\sqrt{5} - \sqrt{2} + 3\sqrt{5} + 4\sqrt{2} = (2 + 3)\sqrt{5} + (-1 + 4)\sqrt{2} = 5\sqrt{5} + 3\sqrt{2}$$

Definition 1.32 Square root

The **square root** $\sqrt{a}$ of a non-negative number a is the non-negative number that gives a when it is squared.

This definition is the whole reason the next rule works. Squaring and square-rooting undo each other, in the same way that multiplying by 2 and dividing by 2 undo each other.

Proposition 1.33 Squaring a square root

For any non-negative number a ,

$$\sqrt{a} \times \sqrt{a} = (\sqrt{a})^2 = a.$$

Multiplying a square root by itself removes the root.

A surd behaves like a variable, except at this step. For a variable, $x \times x = x^2$ and there is nothing more to do. For a surd, $\sqrt{6} \times \sqrt{6} = (\sqrt{6})^2$ collapses to the whole number 6.

Working example a root multiplied by itself

(a) Directly from the rule:

$$\sqrt{6} \times \sqrt{6} = 6, \quad (\sqrt{11})^2 = 11.$$

As a check, $\sqrt{6} \approx 2.449$ and $2.449 \times 2.449 \approx 6$.

(b) With a coefficient, multiply the numbers and the roots separately:

$$5\sqrt{3} \times \sqrt{3} = 5 \times (\sqrt{3})^2 = 5 \times 3 = 15.$$

(c) When the whole term is squared, both the number and the root are squared:

$$(3\sqrt{5})^2 = 3^2 \times (\sqrt{5})^2 = 9 \times 5 = 45.$$

(d) Inside a longer expression, the rule applies term by term:

$$\sqrt{2}(\sqrt{2} - 3) = (\sqrt{2})^2 - 3\sqrt{2} = 2 - 3\sqrt{2}.$$

(e) The rule is what removes a root from a denominator. Multiplying the top and bottom by $\sqrt{3}$ does not change the value:

$$\frac{1}{\sqrt{3}} = \frac{1 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{3}}{3}.$$

Some additional rules for manipulating surds:

Proposition 1.34 Roots of products and quotients

For non-negative numbers a and b (with $b \neq 0$ for the quotient),

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

The same rules hold for cube roots, fourth roots and so on.

The product rule follows from $(\sqrt{a})^2 = a$. Squaring $\sqrt{a}\sqrt{b}$ gives

$$(\sqrt{a}\sqrt{b})^2 = (\sqrt{a})^2 (\sqrt{b})^2 = ab.$$

So $\sqrt{a}\sqrt{b}$ is a non-negative number whose square is ab , which is exactly what $\sqrt{ab}$ means.

Working example combining and splitting roots

- (a) Combining two roots into one can give a whole number:

$$\sqrt{3}\sqrt{12} = \sqrt{3 \times 12} = \sqrt{36} = 6.$$

- (b) The quotient rule works in the same way:

$$\frac{\sqrt{50}}{\sqrt{2}} = \sqrt{\frac{50}{2}} = \sqrt{25} = 5.$$

- (c) Splitting a fraction under a root:

$$\sqrt{\frac{16}{25}} = \frac{\sqrt{16}}{\sqrt{25}} = \frac{4}{5}.$$

- (d) Cube roots follow the same pattern:

$$\sqrt[3]{3}\sqrt[3]{9} = \sqrt[3]{3 \times 9} = \sqrt[3]{27} = 3.$$

- (e) Used in reverse, the product rule simplifies a surd. Split the number into a product where one factor is a perfect square:

$$\sqrt{45} = \sqrt{9 \times 5} = \sqrt{9}\sqrt{5} = 3\sqrt{5}.$$

The rule does not work for addition: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. For example:

$$\sqrt{36+64} = \sqrt{100} = 10, \quad \text{but} \quad \sqrt{36} + \sqrt{64} = 6 + 8 = 14.$$

Another way to simplify surds applies if the number inside the square root sign has a factor that is a perfect square. For example, $\sqrt{72}$ can be simplified because $72 = 36 \times 2$ and 36 is a perfect square. This is due to the fact that:

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

Working example pulling out square factors

Consider the following surds: $\sqrt{80}$ and $\sqrt{15}$.

- (a) 80 has a factor of 16, which is a perfect square. So we can write:

$$\sqrt{80} = \sqrt{16 \times 5} = \sqrt{16}\sqrt{5} = 4\sqrt{5}.$$

- (b) 15 has no factors that are perfect squares (other than 1), so it cannot be simplified. So

we write:

$$\sqrt{15} = \sqrt{15}.$$

As it can't be simplified, it is left in root form.

When simplifying a surd, look for the largest perfect square that is a factor of the number under the root. The first 15 perfect squares are

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225.

Definition 1.35 Rationalising the denominator

Rewriting a fraction so that its denominator no longer contains an irrational root is called **rationalising** the denominator. It is done by multiplying the top and bottom of the fraction by a suitable surd, which does not change the value of the fraction.

When the denominator is a single root, multiply the top and bottom by that root. The rule $(\sqrt{a})^2 = a$ then removes the root from the denominator:

$$\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}.$$

Definition 1.36 Conjugate

A **conjugate** of a two-term expression is the expression obtained by changing the sign of one of its terms, usually the second. For example, a conjugate of $2\sqrt{3} + \sqrt{5}$ is $2\sqrt{3} - \sqrt{5}$, and a conjugate of $5 - 3\sqrt{2}$ is $5 + 3\sqrt{2}$.

Proposition 1.37 Strategy for rationalising the denominator

- If the denominator is a single term containing $\sqrt{a}$, multiply the top and bottom by $\sqrt{a}$.
- If the denominator has two terms involving square roots, multiply the top and bottom by a conjugate of the denominator.

The conjugate works because the new denominator has the form $(A + B)(A - B)$, which is the difference of two squares $A^2 - B^2$. Squaring each term removes its square root.

Working example clearing roots from a denominator

(a) Only the $\sqrt{5}$ needs removing, so multiply by $\sqrt{5}$:

$$\frac{2}{7\sqrt{5}} = \frac{2}{7\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{7 \times 5} = \frac{2\sqrt{5}}{35}.$$

(b) A conjugate of $5 - 3\sqrt{2}$ is $5 + 3\sqrt{2}$:

$$\begin{aligned}\frac{\sqrt{3}}{5 - 3\sqrt{2}} &= \frac{\sqrt{3}}{5 - 3\sqrt{2}} \times \frac{5 + 3\sqrt{2}}{5 + 3\sqrt{2}} = \frac{5\sqrt{3} + 3\sqrt{2}\sqrt{3}}{5^2 - (3\sqrt{2})^2} \\ &= \frac{5\sqrt{3} + 3\sqrt{6}}{25 - 9 \times 2} = \frac{5\sqrt{3} + 3\sqrt{6}}{7}\end{aligned}$$

A final answer that is a surd should be simplified as far as possible. Rationalising the denominator is not always required, so leaving $1/\sqrt{5}$ as it is is acceptable. Many calculators rationalise automatically and display $1/\sqrt{5}$ as $\sqrt{5}/5$.

1.3.3 Powers and the index laws

Definition 1.38 Base and index

In $3^4 = 3 \times 3 \times 3 \times 3 = 81$, the number 3 is the **base** and 4 is the **power**, **index** or **exponent**. The plural of index is **indices**. The result, 81, is also called a power of 3.

Proposition 1.39 Index laws for a single base

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}.$$

The first law extends to more than two powers, for example $a^m a^n a^r = a^{m+n+r}$.

When considering **negative indices**, such as 3^{-3} , we can see that this is the same as the **reciprocal** of 3^3 , which is $\frac{1}{3^3} = \frac{1}{27}$.

Proposition 1.40 Zero and negative index laws for a single base

For any non-zero number a :

- Any power with index 0 equals 1:

$$a^0 = 1.$$

- A negative index means taking the reciprocal of the matching positive power:

$$a^{-n} = \frac{1}{a^n}.$$

- The reverse also holds, so a negative power in a denominator becomes a positive power in the numerator:

$$\frac{1}{a^{-n}} = a^n,$$

$$\text{since } \frac{1}{a^{-n}} = 1 \div \frac{1}{a^n} = a^n.$$

With $n = 1$, this gives $a^{-1} = \frac{1}{a}$, so an index of -1 simply gives the reciprocal.

The index laws above involve just one base number, but there are two more index laws that involve two base numbers.

Proposition 1.41 Index laws for two base numbers

The powers of a product and a quotient can be expressed as follows:

$$(ab)^n = a^n b^n, \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

Working this through an example:

Working example negative indices as a product of fractions

Write each factor as its own fraction. A factor with a negative index flips to the other side of its fraction line and its index becomes positive. Then multiply the fractions back together.

(a) Only the denominator has a negative index:

$$\begin{aligned} \frac{x^3}{x^{-2}} &= \frac{x^3}{1} \times \frac{1}{x^{-2}} = \frac{x^3}{1} \times \frac{x^2}{1} \\ &= x^3 x^2 = x^5 \end{aligned}$$

(b) Raise every factor in the term to the power 3 first, then flip the factor with the negative index:

$$\begin{aligned} (3x^{-2}p)^3 &= 3^3 \times (x^{-2})^3 \times p^3 = \frac{27}{1} \times \frac{x^{-6}}{1} \times \frac{p^3}{1} \\ &= \frac{27}{1} \times \frac{1}{x^6} \times \frac{p^3}{1} = \frac{27p^3}{x^6} \end{aligned}$$

Index laws used. In (a), $\frac{1}{a^{-n}} = a^n$ flips x^{-2} into the numerator, then $a^m a^n = a^{m+n}$ combines $x^3 x^2$ into x^5 . In (b), $(ab)^n = a^n b^n$ applies the power to every factor, $(a^m)^n = a^{mn}$ turns $(x^{-2})^3$ into x^{-6} , and $a^{-n} = \frac{1}{a^n}$ moves x^{-6} into the denominator as x^6 .

1.3.4 Fractional powers